

Peters' Rule Connectivity Engine

Project Documentation

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1 Description

This engine serves as the mathematical heart of the connectivity model. It translates the biological concept of **Peters' Rule**—which states that synaptic connections occur where axons and dendrites physically overlap in space—into a highly efficient, matrix-based algorithm.

Instead of creating a massive, computationally expensive “global grid” encompassing the entire neural circuit, this engine utilizes relative bounding-box mathematics to analyze only the specific pocket of space where two neurons intersect.

2 Core Algorithm: Step-by-Step Breakdown

2.1 1. Loading the Spatial Fields

The function begins by loading the `.npy` files generated in previous pipeline stages:

- **Axonal Field:** The spatial probability distribution of the pre-synaptic neuron (sender).
- **Dendritic Field:** The spatial probability distribution of the post-synaptic neuron (receiver).

At this stage, these are independent 3D arrays of probabilities with their respective somas positioned at the center of their respective arrays.

2.2 2. Calculating the “Voxel Shift”

To compare the two arrays, the physical distance in the brain must be simulated:

1. Calculate the physical vector between somas:

$$physical_shift = pre_soma_pos - post_soma_pos$$

2. Convert physical distance (e.g., $30\ \mu\text{m}$) into discrete array indices based on resolution:

$$voxel_shift = np.round(physical_shift/voxel_size)$$

For instance, if the voxel size is $10\ \mu\text{m}$, a $30\ \mu\text{m}$ shift requires the pre-synaptic array to slide exactly 3 indices to align with the post-synaptic array.

2.3 3. Bounding Box Intersection (The Slicing Trick)

To optimize memory, the function avoids padding arrays with zeros. Instead, it identifies the exact indices where the two boxes intersect.

- If the pre-synaptic grid shifts +3 on the X-axis, the overlap starts at index 3 of the post-synaptic grid and index 0 of the pre-synaptic grid.
- The function generates `slice()` objects for both arrays, creating a “window” into the sub-volume where the two bounding boxes overlap.

2.4 4. Calculating Joint Probability

With the overlapping sub-grids isolated, an element-wise matrix multiplication is performed:

$$P_{joint} = P_{axon} \times P_{dend}$$

This handles the biological logic: if the axon has a 0.5 probability of existence but the dendrite is absent (0.0), the joint probability remains 0.0. Only voxels where both structures possess non-zero probability yield a positive result.

2.5 5. Coordinate Re-mapping

Voxels with $P_{joint} > 0$ must be mapped back to global brain coordinates:

- **Sub-grid to Local:** Add slice starting indices to find the voxel position in the original post-synaptic array.
- **Local to Distance:** Calculate the distance from the array center (soma). A voxel at index 55 (with center 50) is +5 voxels (+50 μm at 10 μm resolution) from the soma.
- **Distance to Global:** Add the relative offset to the absolute global coordinate of the post-synaptic soma to find the real-world 3D centroid.

2.6 6. Probability Normalization

The final array of overlapping centroids is normalized to distribute a specific integer number of synapses:

$$normalized_probabilities = \frac{probabilities}{\sum probabilities}$$

This creates a strict Probability Mass Function (PMF). If a voxel has a value of 0.15, there is a 15% chance any given synapse will be placed in that micro-region.

3 Getting Started

3.1 Prerequisites

- Python 3.8+
- NumPy
- SciPy

3.2 Installation

```
git clone https://github.com/username/peters-rule-engine.git
cd peters-rule-engine
pip install -r requirements.txt
```

4 Usage

```
from engine import calculate_connectivity

# Example execution
synapses = calculate_connectivity(pre_field, post_field, pre_pos, post_pos)
```

5 Contributing

Please submit a Pull Request or open an Issue for any proposed changes. Ensure that all mathematical logic updates are reflected in the documentation.

6 License

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7 Contact

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